

Charles Chow

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SUMMARY

Mechatronics engineer with experience in robotic hardware design, electromechanical motion systems, embedded sensor firmware, CAD, fixtures, BOMs, and hardware/software debugging. Mechanical foundation with CS/ECE graduate training across design, controls, scripting, and prototype testing.

SKILLS

Mechanical / CAD	SolidWorks, NX, Fusion 360, Ansys, FEA, GD&T, BOMs, production drawings, 3D prototyping
Mechatronics	Stepper/servo/DC motors, actuators, sensors, fixtures, test rigs, robotic parts, wiring, integration
Embedded	C/C++, Python, Zephyr, nRF52, BLE, GATT, UART, I2C, SPI, RTOS concepts, Linux
Electrical / Test	KiCad, PCB layout, Gerbers, schematics, soldering, crimping, cable harnessing, multimeter, oscilloscope basics
Systems	Control integration, hardware test plans, reliability testing, failure logs, sensor pipelines, Git, design reviews

RELEVANT EXPERIENCE

Embedded and Software Intern, NeuroLeap

May 2025 – Dec 2025

- Designed embedded C firmware on Nordic nRF52840 for 20 Hz multi-channel BLE sensor streaming, supporting validation, algorithm development, and QA workflows across 5+ engineers.
- Tuned BLE link parameters to reduce packet loss from 9.2% to 1.8%, cutting recalibration time by 60% and enabling uninterrupted 8-hour test sessions.

Mechanical Engineer, Keystone Tower Systems

Jun 2022 – Mar 2023

- Designed electromechanical motion subsystems for a \$2.3M automated welding system using stepper, servo, and DC motor concepts; worked with controls engineers on actuator specs and motion sequencing, achieving 15% faster cycle times.
- Built kinematic layouts enforcing ± 0.5 mm positioning tolerances across 6 configurations; resolved 3 collision scenarios through simulation-driven design review.

Mechanical Engineer, Engineering and Industrial Services

2021 – 2022

- Led design direction for industrial machine fabrication projects, coordinating 6 team members across product, part, and component development.
- Created SolidWorks designs for full-scale industrial machine prototypes, including sheet metal, machined parts, fabricated assemblies, BOMs, and vendor-ready drawings.

SELECT PROJECTS

Witmotion BLE IMU Streaming — C, Python, Zephyr, nRF52, BLE, SEGGER RTT

- Built Zephyr firmware on an nRF52832 DK as a BLE central for a Witmotion IMU, with GATT discovery, notification handling, reconnect scanning, and packet decoding.
- Decoded accel/gyro/orientation packets with CRC16 validation; logged timestamped CSV data with sequence, RSSI, Hz, and IMU fields.

ROS2 Mobile Robot Stack — ROS2, Nav2, Gazebo, Python, Linux

- Built a differential-drive ROS2 robot model in URDF/xacro and Gazebo, maintaining a stable tf2 transform tree.
- Implemented Nav2 waypoint navigation with launch-based bring-up, planner/controller tuning, rosbag2 logging, and diagnostics.

Motor Driver PCB Design — KiCad, Schematic Design, PCB Layout

- Designed a KiCad motor driver PCB for stepper and DC motor control, including schematic capture, component selection, footprints, and design rule checks.
- Produced fabrication-ready Gerbers for motor driver ICs, power regulation, decoupling, and logic-level interfacing.

EDUCATION

Purdue University | M.S. Electrical and Computer Engineering (Part Time, Online)

In Progress

Northeastern University | M.S. Computer Science

May 2026

The Ohio State University | B.S. Mechanical Engineering

Dec 2020